

QA Consistency Sweep — 2026-06-14

SECTION 06-operations-it TYPE document STATUS **active** OWNER Founder UPDATED 2026-08-19

[!NOTE] **Superseded 2026-08-19 — retained as a record, not corrected.** This document captured the position as it stood on its own date. The price card, the B2B pricing model and the platform architecture have all changed since: see [01-governance/decision-log.md](#) **DEC-011** and [03-finance/monetization.md](#) for what is current. Nothing below has been edited — its value is that it records what was decided or observed at the time, including the parts that later turned out to be wrong.

Scope: Issue A (GCP region labelling) and Issue B (corpus-count conflicts) across the full project tree, excluding `node_modules/` and `.git/`. Git worktrees under `flygaca/.claude/worktrees/` are inventoried but treated as non-canonical (they mirror the main repo at older points and are not independently edited).

CRITICAL FINDING — Issue A brief contains an inverted fact

The task brief states: *"me-central2 is Qatar (Doha); the Saudi Arabia (Dammam) region is me-central1"*

This is backwards. The project codebase and GCP geography are correct as written:

GCP Region	Location	Project role
me-central1	Doha, Qatar	Interim compute (Cloud Run not yet granted in Dammam)
me-central2	Dammam, Saudi Arabia	PDPL target — Firestore already here; compute pending

Evidence from the codebase:

- `flygaca/functions/region.js` line 21: `module.exports = 'me-central1'; // interim (Doha); target me-central2 (Dammam)`
- `library/06-product-eng/runbooks/runbook-pdpl-me-central2.md` title and throughout: *"move to me-central2 (Dammam)"; "Compute runs interim in me-central1 (Doha)"*
- `library/06-product-eng/hosting-facts.md`: *"me-central1 (Doha, Qatar) is NOT in-Kingdom";* table lists `me-central2 (Dammam)` for Firestore and Cloud Functions
- Live Firestore database confirmed at `me-central2 (Dammam)` per RUNBOOK checklist
- Cloud Run URL in use: `https://captadel-30479965011.me-central1.run.app` — which is confirmed as the Qatar/Doha interim

The two documents that have the mapping **inverted** (and should be corrected) are:

1. `book-of-fly-gaca-review-2026-06-14.md` **line 60** — states *"me-central2 is Qatar"* (wrong)
2. `.claude/agents/flygaca-qa-reviewer.md` **line 15** — states *"Saudi GCP region is me-central1 = Dammam, not me-central2 = Qatar"* (wrong — has both sides reversed)

No other file in the project has the mapping wrong. The ~170 other `me-central` occurrences consistently and correctly use `me-central2 = Dammam` and `me-central1 = Doha`.

Issue A — GCP Region ("me-central") Occurrences

1A. Inventory

Total files containing `me-central` (excluding `node_modules`, `.git`): **~50 unique canonical files** (plus worktree mirrors). Key non-worktree files:

File	Lines	Context / label used	Correct?
<code>the-book-of-fly-gaca.html</code>	642, 664	<code>me-central2</code> placed in Dammam	✓ Correct
<code>flygaca-resume-briefing-2026-05-23.md</code>	91, 101, 148, 174	<code>me-central2</code> (Dammam)	✓ Correct
<code>01-governance/CLAUDE.md</code>	107, 126-129, 143-144	<code>me-central1</code> (Doha) interim; <code>me-central2</code> (Dammam) target	✓ Correct
<code>library/06-product-eng/setup/setup-vps.md</code>	27, 36	<code>me-central2</code> / Cloud Functions	✓ Correct
<code>library/06-product-eng/setup/setup-firebase.md</code>	15, 18-19	<code>me-central2</code> (Dammam, Saudi Arabia)	✓ Correct
<code>library/06-product-eng/runbooks/runbook-cloudflare.md</code>	5, 33	<code>me-central2</code> target; <code>me-central1</code> interim	✓ Correct
<code>library/06-product-eng/runbooks/runbook-captain-adel.md</code>	55, 107, 207	<code>me-central1</code> for live endpoint (interim Doha)	✓ Correct
<code>library/06-product-eng/runbooks/runbook-captadel-deploy.md</code>	21-22, 94-96, 123-124, 151, 160-161	<code>me-central2</code> Dammam target; <code>me-central1</code> Doha interim	✓ Correct
<code>library/06-product-eng/runbooks/runbook-security-rollout.md</code>	94	<code>me-central1</code> for current function	✓ Correct (reflects live interim)
<code>library/06-product-eng/runbooks/runbook-launch.md</code>	148	<code>me-central2</code> / Dammam	✓ Correct

File	Lines	Context / label used	Correct?
library/06-product-eng/runbooks/runbook-ios.md	68	me-central1 URL (live interim)	✓ Correct
library/06-product-eng/runbooks/runbook-captadel-saas.md	14, 50, 81, 133	me-central2 (Dammam) target	✓ Correct
library/06-product-eng/runbooks/runbook-captadel-extraction.md	129	me-central1 for current Cloud Run	✓ Correct (interim)
library/06-product-eng/runbooks/runbook-pdpl-me-central2.md	all	me-central2 = Dammam; me-central1 = Doha	✓ Correct throughout
library/06-product-eng/hosting-facts.md	17, 24-26, 30-32, 40, 42, 64, 66	me-central2 (Dammam) as canonical	✓ Correct
flygaca-claude-briefing.md	110, 120, 151	me-central2 (Dammam)	✓ Correct
00-strategy/roadmap.md	172, 207, 255	me-central2 for Firestore	✓ Correct
00-strategy/phase0.md	193, 195, 201, 212-215, 234	me-central2 (Dammam)	✓ Correct
master-paperwork-template-index-2026-06-14.md	195	Runbook name reference only	✓ Correct
02-legal/privacy-notice-full-stage-draft-2026-06-14.md	145-146, 189	me-central2 (Dammam, KSA)	✓ Correct
flygaca-antigravity-agents.md	102, 113, 131	me-central2 (Dammam)	✓ Correct
flygaca-phase0-status-2026-05-23.md	81, 109, 121, 135	me-central2 (Dammam)	✓ Correct
flygaca/assistant/captain_adel.py	25, 172	me-central2 as default	✓ Correct
flygaca/functions/region.js	9, 21	me-central1 (Doha) interim; comment names Dammam as target	✓ Correct
flygaca/functions/.env.flygaca-app	throughout	me-central1 for current deploy; names Dammam as target	✓ Correct
flygaca/firebase.json	rewrites	me-central1 (live interim)	✓ Correct
flygaca/privacy.html	115	me-central2 (Dammam)	✓ Correct

File	Lines	Context / label used	Correct?
flygaca/roadmap.md	207	me-central1 (Doha – closest CF v2 region)	✅ Correct
book-of-fly-gaca-review-2026-06-14.md	60, 107	"me-central2 is Qatar" — WRONG	❌ INCORRECT
.claude/agents/flygaca-qa-reviewer.md	15	"Saudi GCP region is me-central1 = Dammam, not me-central2 = Qatar" — WRONG (both sides reversed)	❌ INCORRECT

1B. Files fixed

FIXED: `.claude/agents/flygaca-qa-reviewer.md` (line 15)

- **Before:** correct region facts (Saudi GCP region is me-central1 = Dammam, not me-central2 = Qatar)
- **After:** correct region facts (Saudi GCP region is me-central2 = Dammam; me-central1 = Doha, Qatar – the current interim compute region)
- **Backup created:** `.claude/agents/flygaca-qa-reviewer.md.bak-2026-06-14`

FIXED: `book-of-fly-gaca-review-2026-06-14.md` (lines 60 and 107)

- **Line 60 before:** However, me-central2 is Qatar – the Saudi hosting intent (Dammam) would be me-central1.
- **Line 60 after:** However, this is correct as written: me-central2 IS Dammam (Saudi Arabia) and me-central1 IS Doha (Qatar). The project codebase is accurate throughout. The confusion was in this review document, not the source.
- **Line 107 before:** Verify and correct GCP region – confirm me-central2 vs me-central1 (Qatar vs KSA) across the whole codebase and all docs
- **Line 107 after:** GCP region verified – me-central2 = Dammam (Saudi Arabia, PDPL target, Firestore live here); me-central1 = Doha (Qatar, current compute interim). Codebase is correct; erroneous statements were only in this review doc and the QA agent definition.
- **Backup created:** `book-of-fly-gaca-review-2026-06-14.md.bak-2026-06-14`

All other `me-central` occurrences: **no edits required** — they are internally consistent and factually correct.

Issue B — Corpus Count Conflicts

2A. Authoritative Figures (evidence-based)

Item	Authoritative count	Primary evidence
GACAR Parts	74	<code>library/GACAR-Source-Corpus/</code> = 74 dirs; <code>gacar-index.json</code> count=74; <code>assets/data/parts/</code> = 74 files
Handbooks (topical ebooks)	21	<code>assets/data/ebooks/</code> = 21 HTML files; <code>ebooks-index.json</code> count=21; GACA source: GACAR topical books
GACA Guidance-Manual PDFs	17	<code>library/GACA/Guidance-Manuals/</code> = 17 PDF files (Vol 1–17) — these are source PDFs , not the app ebooks
Aerodromes	61	<code>airports.json</code> count=61; <code>aerodromes-index.json</code> documents=61; <code>assets/data/aerodromes-index.json</code>
VFR Charts	13	<code>assets/data/charts/</code> = 13 image files
Reference documents	190	<code>reference-index.json</code> count=190; <code>_extracted/</code> = 190 TXT files
Tools (flight tools)	35 (HTML files), functionally 24 interactive tools	<code>flygaca/tools/</code> = 35 HTML incl. index + utilities; <code>flygaca.html</code> hero says "24 flight tools"; action plan confirms "24 tool pages exist"
Guides	10 (non-index HTML)	<code>flygaca/guides/</code> = 10 non-index HTML files; action plan: "guide count corrected 11→10"
RAG chunks (current)	47,361	<code>runbook-captain-adel.md</code> (74 Parts + 21 handbooks + 190 ref docs)
RAG chunks (old Book figure)	29,749	Stale — was 74 Parts + 17 eBook PDFs only (pre-handbook integration)

Key distinction: The Book's "17 administration eBooks" refers to the 17 GACA Guidance-Manual PDFs in `library/GACA/Guidance-Manuals/`. The app's "21 handbooks" are the 21 topical HTML ebooks in `assets/data/ebooks/` (a different, larger set derived from the GACA topical books). These are not the same thing. The Book's corpus description predates the handbook integration.

2B. Conflict Table

Every conflicting figure found, with file, line, current value, and recommended correction:

File	Line	Current	Recommended	Notes
the-book-of-fly-gaca.html	771	"96 documents — 75 GACAR Parts, 17 administration eBooks"	"74 GACAR Parts · 21 handbooks · 190 reference documents (+ 61 aerodromes · 13 charts)"	Stale pre-handbook counts. "75" is wrong (74); "17 admin eBooks" confuses Guidance-Manual PDFs with the app's 21 HTML handbooks; "96" arithmetic does not reconcile with any current set
the-book-of-fly-gaca.html	771	"29,749 chunks"	"47,361 chunks"	Stale chunk count pre-dates handbook integration per runbook-captain-adel.md
library/06-product-eng/runbooks/runbook-launch.md	18–19	"17 Flight Tools, 7 Guides"	"24 flight tools, 10 guides" (or note as pre-launch snapshot)	Stale; live site and action plan confirm 24 tools, 10 guides
flygaca/office/runbook-launch.md	18–19	"17 Flight Tools, 7 Guides"	Same as above	Mirror of library copy; same correction needed
book-of-fly-gaca-review-2026-06-14.md	53–55	Quotes the "96 documents — 75 GACAR Parts, 17 eBooks" and calls it conflicting	Remains accurate as a review observation — but the resolution is now documented here	No edit needed (it's noting the conflict)

2C. Figures that are consistent across authoritative sources

The following numbers appear identically in the live site (flygaca.html , pricing.html , library.html), the ROADMAP, the action plan, the RUNBOOK-captain-adel, and the product-profile:

- **74 GACAR Parts** — consistent everywhere except The Book (says 75)
- **21 handbooks** — consistent everywhere except The Book (says 17)
- **61 aerodromes** — consistent everywhere
- **13 charts** — consistent everywhere
- **190 reference documents** — consistent everywhere
- **24 flight tools** — consistent in live site; RUNBOOK-launch is stale at 17
- **10 guides** — consistent in live site; RUNBOOK-launch is stale at 7

2D. Fixes applied

No mass-edits made to prose documents. The two stale-count files require human confirmation before editing:

Recommended human edits (not auto-fixed):

1. `the-book-of-fly-gaca.html` **line 771** — Change "96 documents — 75 GACAR Parts, 17 administration eBooks, and the references that bind them. It is rebuilt by `build_library.py` and indexed by BM25 across **29,749 chunks**." to reflect current figures. Suggested: *"The corpus spans 74 GACAR Parts, 21 topical handbooks, 190 reference documents, 61 aerodromes and 13 charts — indexed by BM25 across **47,361 chunks**."* The "96 documents" framing is both wrong and confusing (the actual total of Parts + handbooks + references alone is 285).
2. `library/06-product-eng/runbooks/runbook-launch.md` **line 18** — Change "17 Flight Tools, 7 Guides" to "24 flight tools, 10 guides" to match the live site. (This is a launch runbook snapshot; the discrepancy is a maintenance concern, not a live-site error.)
3. `flygaca/office/runbook-launch.md` **line 18** — Same correction as above (this is a mirror of the library copy).

Summary of Files Edited vs Flagged

Files edited (with backups)

File edited	Backup	Change
<code>/sessions/nice-quirky-pasteur/mnt/Fly GACA / .claude/agents/flygaca-qa-reviewer.md</code>	<code>.bak-2026-06-14</code> created	Fixed inverted region labels on line 15
<code>/sessions/nice-quirky-pasteur/mnt/Fly GACA /book-of-fly-gaca-review-2026-06-14.md</code>	<code>.bak-2026-06-14</code> created	Corrected inverted region claim at lines 60 and 107

Files flagged for human decision

File	Issue	Recommended action
<code>the-book-of-fly-gaca.html</code>	Line 771: "96 documents — 75 GACAR Parts, 17 admin eBooks, 29,749 chunks" — all three figures stale	Human to update to 74 Parts · 21 handbooks · 190 refs / 47,361 chunks
<code>library/06-product-eng/runbooks/runbook-launch.md</code>	Line 18: "17 Flight Tools, 7 Guides" — stale launch snapshot	Human to update to 24 tools / 10 guides
<code>flygaca/office/runbook-launch.md</code>	Line 18: same stale figures	Same correction as above

Files with correct me-central labelling (no edits needed)

All remaining ~48 files containing `me-central` — including all source code, runbooks, briefings, privacy notice, PHASE0, ROADMAP, and CLAUDE.md — correctly map `me-central2` = Damman (Saudi Arabia, PDPL target) and `me-central1` = Doha (Qatar, interim compute).

Region Truth Table (for reference)

Claim	Verdict
GCP <code>me-central1</code> = Doha, Qatar	✓ CORRECT
GCP <code>me-central2</code> = Dammam, Saudi Arabia	✓ CORRECT
Firestore <code>(default)</code> is in <code>me-central2</code> (Dammam)	✓ CONFIRMED (RUNBOOK checklist, privacy.html)
Cloud Functions / Cloud Run are in <code>me-central1</code> (Doha)	✓ CONFIRMED INTERIM — pending Google account access grant for me-central2 compute
The PDPL target for compute is <code>me-central2</code> (Dammam)	✓ CORRECT (region.js comment, RUNBOOK-pdpl-me-central2)

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